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Introduction

The Fire and Security Alarm Monitoring System (FASAM) plan which was developed prior to this needs a risk assessment to be conducted. This report will focus on a risk management strategy which aims to identify and explain potential risks that could arise, their likelihood, and impact. In addition to that several matrixes will be presented including quantitative assessment of goals requirements, risk countermeasures matrix, risk reduction strategy, KRI matrix along with relevant calculations which will be conducted using the formulas. The report will be divided into 4 stages.

Stage 1: Risk Identification assessment

Stage 2: Risk analysis and risk matrix

Stage 3: Risk countermeasures matrix and analysis

Stage 4: Key Risk indicators (KRIs)

Stage 1: Risk Identification Assessment

Risk is defined as the probability of an adverse influence with the magnitude of its severity. (Liu and Jiang, 2012) It is measured using the following formula:

$\text{Risk} = \text{Probability} \times \text{Impact}$.

Risk management is an extremely important process to reduce and prevent the likelihood of risks making a large impact. Risk management is the application of management, policies, procedures and practices to identify, analyse, control, and prevent risk. (Bartoo, 2003).

The risk management strategy begins by identifying potential risks in the Fire and Security Alarm Monitoring System (FASAM). A risk identification matrix has been developed, all identified risks will be assigned to a unique Risk ID, their likelihood and impact will be rated from 1 to 5, 1 being the lowest risk and 5 being the highest level of risk. In addition to that each of these risks will be added into a low, medium or high category based on the risk level. These risks will have clear descriptions about their possible consequences. Risks with medium to high level will be analysed further in stage 2 for quantitative assessment and calculations.

ID	Risk	Description	Likelihood	Impact	Level
1	Fire alarm failure	Fire alarm not working when the Central control area is unmanned	2	5	10 (Med)
2	Fire alarm failure	System is not able to detect smoke, heat and fire	3	5	15 (Med)
3	Sensor failure	System sensors fail to measure the intensity of smoke, heat and fire	3	4	12 (Med)
4	System diagnostics failure	System fails to perform automatic diagnostics to check if the sensors are operational and working properly.	3	4	12 (Med)
5	False alarm detection failure	System does not detect and stop a false alarm	3	4	12 (Med)

6	Fire system failure	System fails to notify central control area about fire incidents and false alarms	2	5	10 (Med)
7	Unable to contact emergency services	Fire system fails to contact emergency services automatically upon confirming a fire alarm.	2+	5	10 (Med)
8	Location sharing failure	System fails to share precise location of fire with emergency services and system operators in central control area.	3	5	15 (Med)
9	Not able to display real time data	The system fails to display real-time data and status of all fire and security alarms and sensors.	3	5	15 (Med)
10	Data storage failure	System fails to save and store data	2	3	6 (Low)
11	Sprinkler system malfunction	Sprinkler system activates when people are present in the room.	2	5	10 (Med)
12	Sprinkler system malfunction	Sprinkler system does not activate when the room is empty, which can lead to the fire becoming more extreme and dangerous	3	5	15 (Med)
13	Electrical system shutdown malfunction	Electrical equipment shuts down when people are present in the room, which can make the fire alarm inaudible to people	2	5	10 (Med)
14	Electrical system shutdown malfunction	Electrical system does not shut down when people are not present in the room which could damage the equipment and make the fire more dangerous as it can cause more damage to the building.	3	5	15 (Med)
15	Audible evacuation instructions malfunction	Audible evacuation warning will stop working if electrical equipment shuts down when people are present in the room. In addition to that, a faulty audible warning system may not work in	4	5	20 (High)

		emergency which will prevent providing evacuation instructions to people present in the building			
16	Exit indicators malfunction	Exit direction indicators may not work if electrical equipment shuts down or if the indicators become faulty.	3	5	15 (Med)
17	Security System Malfunction	Security system fails to lock doors, and it is not able to isolate zones from the building.	3	5	15 (Med)
18	Risk of network delay	Vulnerability in the network, which prevents the system to respond quickly. (Ganguly and Mansouri, 2011)	4	5	20 (High)
19	Software performance failure	The software system might slow down and stop working	3	5	15 (Med)
20	Risk of attacks from hackers	Malicious attacks on fire and security system which could compromise the data. (Abbass, Baina and Bellafkih, 2015)	5	5	25 (High)
21	Risk of excessive power consumption	System fails to activate power saving mode.	2	1	2 (Low)
22	Slow UI dashboard	The UI dashboard of system becomes slow with minor delays	3	2	6 (Low)
23	Hardware system malfunction	Hardware system can overheat, become faulty, slow down or may get damaged including damage due to human factors	4	5	20 (High)
24	Wrong language installation	System may install wrong language, although default language of the system is English, when users configure system with other languages it may not complete that task and the language is set to default as English.	2	1	2 (Low)

Risk level criteria:

A special criterion will be created to assess the level of risks. Risk level was calculated by using the formula $\text{Risk} = \text{Probability} \times \text{Impact}$ ratings were collected. The rating for different risks varies. Lowest rating is 2, while the highest rating is 25. Highest possible level can be 25 if both the likelihood and impact are 5.

Risk categories are divided into following level:

Low risk level = 0 to 9 (Green)

Medium risk level = 10 to 19 (Amber)

High risk level = 20 to 25 (Red)

Risks with lower impact and likelihood had low ratings, the ratings increased as the level of impact and likelihood increased.

Justification for ratings:

In risk 1 Fire alarm failure's likelihood rating is 2 because system will be well maintained, regular checks on fire alarm will be conducted, however if the fire alarm fails, the impact will be extremely high as it can risk and cause harm to human life. However, the chances are higher for sensors failing to detect smoke, heat and fire, this is due to with time sensors could get old and become faultier and inaccurate and its ratings is 3. System could fail to notify central control area about fire incidents, however due to intensive planning of the development of system, the chances of this will be low which reflects the rating 2, however its impact will be very high, and the rating is 5. Audible warning system can fail, and the system could face network issues, the level of these risks has been rated high because these situations have high chance of occurring.

With more safety critical systems relying on the internet and cloud services, attackers are targeting these high value targets, the fire and security system can be at a high risk from an attack from hackers (Mhara, Abdulrahman and Baroud, 2024), that is why the rating of 5 is given for both impact and likelihood. Other risks such as power saving failure and wrong language installation have low likelihood and impact, so they have been given lower ratings.

Stage 2: Goals identification and risk matrix

The risks with medium to high level are further analysed for quantitative risk assessment. The goals which will be added will aim to ensure that the system is able to fulfil its requirements and for the purpose it will be designed for.

Goal A: Quick and fast emergency response by the system

Goal B: Reliable and highly maintained system infrastructure and software system

Goal C: Secure and fast fire and security system

Goal D: Human safety during evacuations

Goal E: Reducing chances of false alarms

Impact of risks on goals are rated from 1 to 10 along with Combined risk criticality and total effect on assets calculations.

Combined risk criticality will be measured by multiplying effect of risk on goals with risk level, higher the risk criticality, the more it will affect goals.

For example, risk level of risk ID 1 is 10, so the calculations are:

$$(10 \times 10) + (9 \times 10) + (8 \times 10) + (10 \times 10) + (1 \times 10) \\ = 100 + 90 + 80 + 100 + 10 = \mathbf{380}$$

Total detrimental effect on assets will be measured by multiplying impact of each risk with risk on goals.

For example, Impact on risk ID 1 is 5, so the calculations are:

$$(10 \times 5) + (9 \times 5) + (8 \times 5) + (10 \times 5) + (1 \times 5) = \\ 50 + 45 + 40 + 50 + 5 \\ = \mathbf{190}$$

ID	Goal A	Goal B	Goal C	Goal D	Goal E	Combined risk criticality	total detrimental effect on asset
1	10	9	8	10	1	$(10 \times 10) + (9 \times 10) + (8 \times 10) + (10 \times 10) + (1 \times 10)$ $= 100 + 90 + 80 + 100 + 10 = \mathbf{380}$	$(10 \times 5) + (9 \times 5) + (8 \times 5) + (10 \times 5) + (1 \times 5)$ $= 50 + 45 + 40 + 50 + 5 = \mathbf{190}$
2	8	9	8	10	1	540	180
3	8	9	7	7	6	480	148
4	8	9	7	8	1	396	132
5	3	7	6	3	10	348	116
6	7	7	6	6	9	350	175
7	10	10	10	10	1	401	225
8	8	8	8	8	1	495	165
9	4	9	6	6	6	375	155
11	4	9	8	10	1	320	160
12	10	10	7	10	1	570	340
13	4	9	9	10	1	330	165
14	7	9	9	5	1	465	155
15	10	10	10	10	1	820	225
16	10	10	10	10	1	615	225
17	10	10	10	9	1	600	200
18	10	10	10	9	1	800	200
19	10	10	9	8	1	570	190
20	9	9	7	7	1	825	165
23	8	10	9	9	4	780	200

Stage 3: Risk Countermeasures

Type of Risk	Countermeasure	Effectiveness (0-1)	Combined effect of countermeasures	Combined risk reduction (Probability)
Fire alarm and Sensor failures. Risk ID:	Regularly conduct maintenance on sensors and change them after certain time.	0.8	$(5 \times 0.8) + (5 \times 0.8) + (4 \times 0.8) = 7.2$	$1 - (0.5 \times 0.5 \times 0.4 \times 0.8) = 0.92$
False Alarms. Risk ID:	Operators in Central Control Area need to be more involved when monitoring the system to prevent false fire alarms manually.	0.6	$4 \times 0.6 = 2.4$	$1 - (0.6 \times 0.4) = 0.76$
Network Vulnerability and security risks. Risk ID:	All sensors, fire alarms, hardware system must be connected to network via wired connection. Cabled communication protocol known as (CAN) can be used. (Antunes et al., 2019) Install firewall and malware detection system.	0.8	$(5 \times 0.8) + (5 \times 0.8) = 8$	$1 - (0.8 \times 0.5 \times 0.5) = 0.8$
Hardware system failure. Risk ID:	Clean hardware regularly. Use efficient cooling equipment. Follow correct shutdown and restart procedure. (Lang, 2024)	0.7	$(3 \times 0.7) + (5 \times 0.7) + 3.5 + 3.5 + 3.5 + 3.5 + 3.5 + 3.5 + 3.5 + 3.5 + 3.5 = 2.1 + 31.5 = 33.6$	$1 - (0.7 \times 0.5^9) = 0.99$

Fire management software issues. Risk ID:	Regular software updates and mock tests on the software systems.	0.7	$(4 \times 0.7) + (4 \times 0.5) + (4 \times 0.5) + (4 \times 0.5) = 10.8$	$1 - (0.7 \times 0.4 \times 0.5^4) = \mathbf{0.98}$
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Stage 4: Key Risk Indicators

(Rasshyvalov and Rushkovskyi, 2022)'s report explains the meaning of Key Risk Indicators (KRI). KRIs are used to monitor the identified risks over a period of time, KRIs are also used to measure impact of a risk.

The KRIs for this project are:

1. Fire alarm failure rate:

Description:

This will measure the rate of fire alarms not responding in real emergency situations.

Formula:

$$(\text{Rate of failed fire alarms} / \text{Total fire alarm incidents}) \times 100$$

Unit measurement:

%

2. Emergency response time:

Description:

This will measure the average time taken by the system to identify and respond to an emergency.

Formula: Record time taken to respond to emergencies. Add all the values of times and divide it by the total figures recorded to find the average.

Unit measurement:

Time (seconds)

3. Security system failure rate:

Description:

This will measure the failure rate for security system during emergency situations including door lock failure to isolate zones, sprinkler system failure etc.

Formula:

$$(\text{Failed security system incidents} / \text{Total security system operations}) \times 100$$

Unit measurement:

%

4. Sensor efficiency:

Description:

This will measure how efficient the sensors are at detecting smoke, fire and heat including their intensity and development of hazards.

Formula:

$$(\text{Number of correct sensor identifications} / \text{total sensor identifications}) \times 100$$

Unit measurement:

%

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